

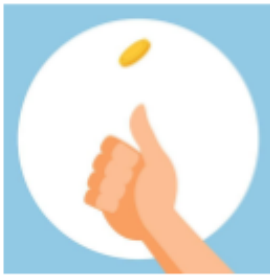
■ Ch 6. Probability Distributions

- Random variables – Discrete
 - Continuous
- Discrete Probability distributions – Binomial, Poisson, Geometric
- Continuous Probability distributions – Uniform, Exponential, Normal

Random Variables

A random variable (r.v.) is a variable that has numerical values which is defined on or determined by the outcomes or events of an experiment.

1. Define RV



$X = \# \text{ Heads in 2 tosses}$

2. Identify sample space

Toss outcomes	X
(T, T)	0
(H, T), (T, H)	1
(H, H)	2

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3. Assign probabilities

→ $\frac{1}{4}$
→ $\frac{1}{2}$
→ $\frac{1}{4}$

Random Variables

Random variables can be discrete or continuous.

Eg: The no. of seeds germinating from a flower pot. Possible values for X are 0, 1, 2, ... (discrete).

The maximum daily temperature in Singapore. Possible values are 23 – 35°C, e.g. 26.1276°C (continuous).

The response to questions with “Yes”, “No”, or “Don't know” answers. This are not a r.v. (the values are not numerical).

Let Y be the number of “Yes”s. Y is a discrete r.v.

Random Variables

Expected Value of a r.v.

For a given set of observed data $\{x_1, x_2, \dots, x_n\}$, one of the central tendency measure is the arithmetic average:

$$\bar{X} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$\bar{X}$ is calculated based on a sample of n observed values.

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$\bar{X}$ is calculated based on a sample of n observed values.

If we know the probability of occurring associated with each value of a r.v., then we can obtain the weighted average which is known as the expectation of the r.v.

Random Variables

Expected Value of a r.v.

The probability distribution or probability mass function (pmf) of a r.v. X is given as:

Values of X	x_1	x_2	$\dots$	x_k
Probabilities	p_1	p_2	$\dots$	p_k

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Its expected value is defined as:

$$E(X) = x_1 p_1 + x_2 p_2 + \dots + x_k p_k = \sum_{i=1}^k x_i p_i$$

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Also denoted by μ_x

Random Variables

Eg: (Decision Analysis)

An oil exploration company has a lease for which it must decide on one of the following options:

1. Sell now and can get \$125K.

Random Variables

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An oil exploration company has a lease for which it must decide on one of the following options:

1. Sell now and can get \$125K.
2. Hold for a year and if oil prices rise (prob = 0.6) it can sell for \$300K or if oil prices fall (prob = 0.4) it can get \$100K.

Random Variables

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3. Drill now. The cost of drilling is \$200K and drilling will lead to one of the following outcomes:

Well type	Dry	Wet	Gusher
Probability	0.5	0.4	0.1
Profit	\$0	\$400K	\$1500K

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Well type	Dry	Wet	Gusher
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Profit	\$0	\$400K	\$1500K

What should the company do?

Random Variables

Eg: (Decision Analysis)

An oil exploration company has a lease for which it must decide on one of the following options:

1. Sell now and can get \$125K.

Financial gain = \$125K

Random Variables

Eg: (Decision Analysis)

An oil exploration company has a lease for which it must decide on one of the following options:

2. Hold for a year and if oil prices rise (prob = 0.6) it can sell for \$300K or if oil prices fall (prob = 0.4) it can get \$100K.

X	300	100
Probability	0.6	0.4

$$E(X) = 300 \times 0.6 + 100 \times 0.4 = \$220K$$

Expected financial gain = \$220K

Random Variables

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$$X = \text{profit} - \text{drilling costs } (\$200\text{K})$$

Random Variables

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An oil exploration company has a lease for which it must decide on one of the following options:

3. Drill now. The cost of drilling is \$200K and drilling will lead to one of the following outcomes:

X	0 – 200K	400K – 200K	1,500K – 200K
Probability	0.5	0.4	0.1

$$E(X) = -200 \times 0.5 + 200 \times 0.4 + 1300 \times 0.1 = \$110K$$

Expected financial gain = \$110K

Random Variables

Option 1 (sell now):

Financial gain = \$125K

Option 2 (hold for 1 yr):

Expected financial gain = \$220K

Option 3, X = profit – drilling costs (\$200K)

Expected financial gain = \$110K

Best decision is to hold for a year and then sell.

Given the pmf of X below, if $Y=(X-1)^2$, find $E[Y]$.

x	0	1	2
$P_X(x)$	0.2	0.5	0.3

Random Variables

Variance of a r.v.

Variance is a measure of variability.

For a sample of n independent observations the sample variance is obtained using:

$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{X})^2}{n-1} \quad \text{or} \quad \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right]$$

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The variance of a r.v. X is defined as

$$\begin{aligned} \text{var}(X) &= p_1(x_1 - \mu)^2 + p_2(x_2 - \mu)^2 + \cdots + p_n(x_n - \mu)^2 \\ &= \sum_{i=1}^n p_i(x_i - \mu)^2 = E[(x - \mu)^2] \end{aligned}$$

where p_i = probability that x_i occurs.

Variance of a r.v.

It represents the theoretical limit of the sample variance s^2 as the sample size n becomes very large.

$\text{Var}(X)$ is often denoted by σ^2 .

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$\text{Var}(X)$ is often denoted by σ^2 .

An alternate formula for $\text{var}(X)$ is as follows:

$$\begin{aligned}\text{var}(X) &= \sum p_i (x_i^2 - 2x_i\mu + \mu^2) \\ &= \sum p_i x_i^2 - 2\mu \sum p_i x_i + \mu^2 \sum p_i \\ &= E[X^2] - \mu^2 \quad \text{or} \quad E[X^2] - (E[X])^2\end{aligned}$$

Random Variables

Eg

Let the random variable X be the number of male students in a group from a class of size 5.

The probability distribution of X is

X	0	1	2	3	4	5
$P(X = x)$	1/32	5/32	10/32	10/32	5/32	1/32

What is the expected number of males $E(X)$ in the group, and what is the standard deviation, σ , of X ?

Random Variables

Solution

$$E(X) = \sum_{i=1}^n x_i P(x_i)$$
$$= 0 \times \frac{1}{32} + 1 \times \frac{5}{32} + 2 \times \frac{10}{32} + 3 \times \frac{10}{32} + 4 \times \frac{5}{32} + 5 \times \frac{1}{32}$$
$$= \frac{80}{32} = 2.5 = \mu$$

i.e. on average such groups have 2.5 males.

Random Variables

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i.e. on average such groups have 2.5 males.

$$\begin{aligned}\text{var}(X) &= \sum x^2 P(x) - \mu^2 \\ &= \left(0^2 \times \frac{1}{32} + \dots + 5^2 \times \frac{1}{32}\right) - (2.5)^2 \\ &= 7.5 - 2.5^2 = 1.25\end{aligned}$$

So $\sigma = \sqrt{\text{var}(X)} = \sqrt{1.25} \approx 1.12$

Given the pmf of X below, if $Y=(X-1)^2$, find $\text{Var}[Y]$.

x	0	1	2
$P_X(x)$	0.2	0.5	0.3

Random Variables

Empirical quantity (m observed data)	Theoretical quantity (mathematical d.r.v.)	Remarks
Mean: $\bar{X} = \sum_i \frac{f_i}{m} x_i$	Expectation: $E[X] = \mu = \sum_i p_i x_i$	

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Relative freq of x_i is $\frac{f_i}{m}$	$P[X = x_i] = p_i$	$\frac{f_i}{m} \rightarrow p_i$ as $m \rightarrow \infty$
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Mean: $\bar{X} = \sum_i \frac{f_i}{m} x_i$	Expectation: $E[X] = \mu = \sum_i p_i x_i$	$\bar{X} \rightarrow \mu$ as $m \rightarrow \infty$
Variance: $s^2 = \sum_i \frac{(x_i - \bar{X})^2 f_i}{m - 1}$	Var[X]: $\sigma^2 = \sum_i (x_i - \bar{X})^2 p_i$	$s^2 \rightarrow \sigma^2$ as $m \rightarrow \infty$

Random Variables

Expected Value & Variance for a Function

(Linear transformation of r.v.)

If $Y = a + bX$ where X is a r.v. and a and b are known constant values, then

$$E(Y) = a + b E(X)$$

and

$$\text{var}(Y) = b^2 \text{var}(X)$$

Random Variables

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$$E(Y) = a + b E(X)$$

and

$$\text{var}(Y) = b^2 \text{var}(X)$$

so

$$\sigma_y = \sqrt{b^2 \text{var}(X)} = \sqrt{b^2 \sigma_x^2} = b \sigma_x$$

Random Variables

Similarly if $T = a + bX + cY$ where X and Y are uncorrelated r.v. and a , b and c are known constants, then

$$E(T) = a + bE(X) + cE(Y)$$

and

$$\text{var}(T) = b^2 \text{var}(X) + c^2 \text{var}(Y)$$

Random Variables

Eg: A company makes products for local and export markets. The number of sales for next year are estimated as follows:

local, X units	1,000	3,000	5,000	10,000
probability	0.1	0.3	0.4	0.2

export, Y units	300	500	700
probability	0.4	0.5	0.1

Random Variables

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Suppose the company makes a profit of \$2000 on each unit sold on the local market and \$3500 on each exported unit. What is the total expected profit?

Random Variables

local, X units	1,000	3,000	5,000	10,000
probability	0.1	0.3	0.4	0.2

First, $E(X) = 1000 \times 0.1 + 3000 \times 0.3 +$
 $5000 \times 0.4 + 10000 \times 0.2$
 $= 5000$ (= expected local sales)

Random Variables

export, Y units	300	500	700
probability	0.4	0.5	0.1

$$\begin{aligned} E(Y) &= 300 \times 0.4 + 500 \times 0.5 + 700 \times 0.1 \\ &= 440 \quad (= \text{expected export sales}) \end{aligned}$$

Random Variables

The company makes a profit of \$2000 on each unit sold on the local market and \$3500 on each exported unit.

Hence the total profit is $T = 2000X + 3500Y$

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Hence the total profit is $T = 2000X + 3500Y$

$$\begin{aligned} E(T) &= 2000E(X) + 3500E(Y) \\ &= 2000 \times 5000 + 3500 \times 440 \\ &= \$11,540,000 \end{aligned}$$

$\Rightarrow$ this is next year's estimated profit.

Random Variables

Eg: A component is made by cutting a metal piece to length X and trimming it by amount Y . Both processes are somewhat imprecise.

Nett length $T = bX + cY$ with $b = 1$ and $c = -1$.

$$E(T) = bE(X) + cE(Y) = E(X) - E(Y)$$

$$\text{Var}(T) = b^2 \text{var}(X) + c^2 \text{var}(Y)$$

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↑
Note

i.e. $\text{var}(T)$ is greater than either $\text{var}(X)$ or $\text{var}(Y)$, even though $T = X - Y$, because both X and Y contribute to the variability in T .

Prob. Distribution of Discrete r.v.

Special Probability Distributions

- Binomial Distribution
- Poisson Distribution
- Geometric Distribution

Prob. Distr. of Discrete r.v.

The list of possible values that a d.r.v. X can take and their probabilities is called the discrete probability distribution (also known as probability mass function, pmf) for X .

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Eg:

Let the random variable X be the number of girls in a family of 3 children.

Possible values:

$X = 3$	<i>GGG</i>		
$X = 2$	<i>GGB</i>	<i>GBG</i>	<i>BGG</i>
$X = 1$	<i>BBG</i>	<i>BGB</i>	<i>GBB</i>
$X = 0$	<i>BBB</i>		

Prob. Distr. of Discrete r.v.

Assume that the 8 outcomes are equally likely,

x_i	0	1	2	3
Probability Distribution $P(X=x_i)$	1/8	3/8	3/8	1/8
Cumulative Distribution $P(X \leq x_i)$	1/8	4/8	7/8	1

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In general, we write:

$$P(X = x_i) = p_i \text{ for } i = 1, \dots, k, \text{ and } 0 \leq p_i \leq 1$$

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Notation convention: we often use capital letters for random variables and small letters for specific values.

Prob. Distr. of Discrete r.v.

Eg: Let X denote the sum of the results of 2 dice thrown.

Outcomes:

1,1	2,1	3,1	...	6,1
1,2	2,2	3,2	...	6,2
⋮	⋮	⋮		⋮
1,6	2,6	3,6	...	6,6

36 outcomes and the values of X are 2, 3, ..., 12.

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⋮	⋮	⋮		⋮
1,6	2,6	3,6	...	6,6

36 outcomes and the values of X are 2, 3, ..., 12.

Assuming that the outcomes are equally likely, i.e. the probability of each outcome is $1/36$, the probability distribution of X is

X	2	3	4	...	10	11	12
$P(X = x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	...	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Prob. Distribution of Discrete r.v.

Special Probability Distributions

- Binomial Distribution
- Poisson Distribution
- Geometric Distribution

Prob. Distr. of Discrete r.v.

Binomial Distribution

- Consider n Bernoulli trials, where each trial is an “experiment” with exactly 2 possible outcomes, “success” and “failure”.
- Assume that the probability of success (S) is the same for all trials, $P(S) = p$, and the probability of failure $P(F) = 1 - p$.
- Assume also that trials are independent, i.e., the probability for any given combination of successes and failures can be obtained by multiplying the probabilities for each trial outcome.

Prob. Distr. of Discrete r.v.

Eg: Consider 5 trials, the probability,

$$\begin{aligned}P(SSFSF) &= p \cdot p \cdot (1-p) \cdot p \cdot (1-p) \\ &= p^3(1-p)^2\end{aligned}$$

The probability of obtaining any 3 successes and 2 failures in 5 trials, i.e. $SSSFF$, $SSFSF$, ... etc., is $p^3(1-p)^2$ for each of the different ways this could occur.

Prob. Distr. of Discrete r.v.

The number of distinct "arrangements" of 3 successes and 2 failures can be calculated using the binomial coefficient $\binom{n}{x}$ or nC_x .

The binomial coefficient is defined as: ${}^nC_x = \frac{n!}{x! (n-x)!}$
where $n! = n \times (n-1) \times \dots \times 2 \times 1$

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In this example, $\binom{5}{3} = \frac{5!}{3!2!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{(3 \times 2 \times 1)(2 \times 1)} = 10$

so there are 10 distinct ways to obtain 3 successes in 5 trials, with each arrangement having a probability of $p^3(1-p)^2$.

Prob. Distr. of Discrete r.v.

Let X be the r.v. equal to the total number of successes in n trials. To calculate the probability of obtaining x successes, we have:

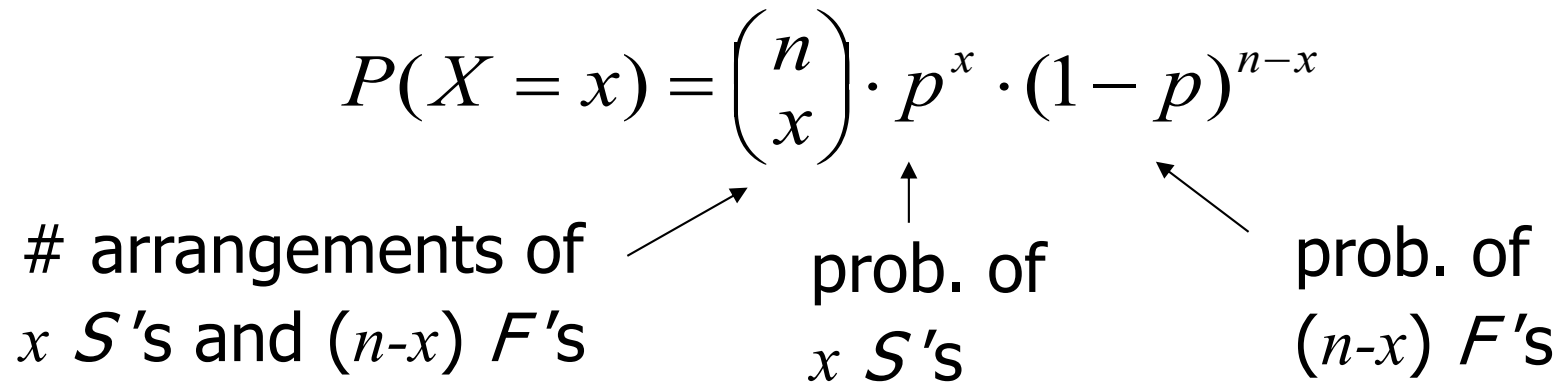
$$P(X = x) = \binom{n}{x} \cdot p^x \cdot (1 - p)^{n-x}$$

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↑
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The distribution of the count of successes is called the binomial distribution with two parameters, n and p .

We say $X \sim B(n, p)$.

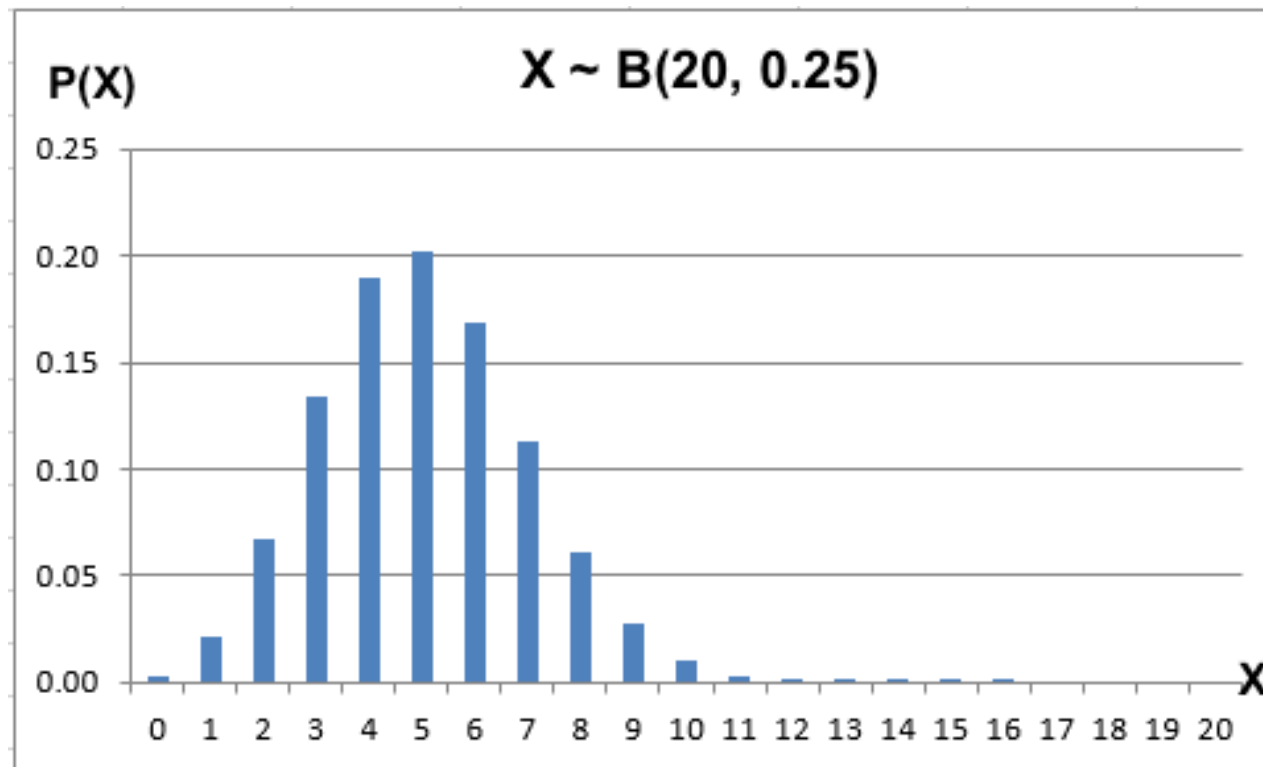
X	0	1	...	n
$P(X=x)$	${}^nC_0 p^0 (1-p)^n$	${}^nC_1 p^1 (1-p)^{n-1}$	...	${}^nC_n p^n (1-p)^{n-n}$

Prob. Distr. of Discrete r.v.

Binomial Distribution – $X \sim B(n, p)$

$$P(X = x) = \binom{n}{x} \cdot p^x \cdot (1 - p)^{n-x}$$

Example:



Prob. Distr. of Discrete r.v.

Mean and variance of $X \sim B(n, p)$

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

The mean of X is given by:

$$E[X] = \sum_{x=0}^n x \binom{n}{x} p^x (1 - p)^{n-x}$$

Recall:

$$E[X] = \sum x p(x)$$

Prob. Distr. of Discrete r.v.

Mean and variance of $X \sim B(n, p)$

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The mean of X is given by:

$$E[X] = \sum_{x=0}^n x \binom{n}{x} p^x (1 - p)^{n-x} = np$$

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And the variance is:

$$Var[X] = E[X^2] - E[X]^2$$

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Recall:
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And the variance is:

$$Var[X] = E[X^2] - E[X]^2 = np(1 - p) = npq$$

where $q = (1 - p)$

Prob. Distr. of Discrete r.v.

Example:

A football team plays 3 games. Assume each game is a Bernoulli trial with $\text{prob}(\text{win}) = 0.5$. Let the r.v. X be the number of wins. What is the probability that the team will win exactly 2 games?

Prob. Distr. of Discrete r.v.

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Solution

X has binomial distribution with $n = 3$ and $p = 0.5$, probability of winning each game. i.e.

$$X \sim B(3, 0.5)$$

Prob. Distr. of Discrete r.v.

Using the formula for Binomial probabilities:

$$\begin{aligned}P(X = x) &= \binom{n}{x} p^x (1 - p)^{n-x} \\&= \binom{3}{2} 0.5^2 (1 - 0.5)^1 \\&= 3/8\end{aligned}$$

Go to **wooclap.com** and use the code **QTINLE**



There are 10 blue and 30 green balls in a bag. A ball is randomly picked, the colour is noted and it is put back into the bag. This process is repeated 10 times. Let X be the number of times a blue ball was picked. Calculate $E[X]$ and $\text{Var}[X]$.



wooclap



100 %



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Prob. Distr. of Discrete r.v.

Example:

A quality control system selects a sample of 10 items from each batch of products for testing.

If 2 or more of the items are defective the whole batch is rejected.

If the probability of an item being defective is 0.05, what is the probability of

- (i) having 2 defectives in the sample?
- (ii) the batch being rejected?

Prob. Distr. of Discrete r.v.

Soln: Let X be the number of defectives in the sample of $n = 10$ items. $X \sim B(10, 0.05)$

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$$(i) P(X=2) = {}^{10}C_2 (0.05)^2 (0.95)^8 = 0.0746 \text{ (to 3.d.p.)}$$

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$$\begin{aligned}(ii) P(\text{reject batch}) &= P(X \geq 2) \\ &= 1 - P(X < 2) \\ &= 1 - P(X=0) - P(X=1) \\ &= 1 - 0.5987 - 0.3151 \\ &= 0.0861 \text{ (to 3.d.p.)}\end{aligned}$$

Prob. Distribution of Discrete r.v.

Special Probability Distributions

- Binomial Distribution
- Poisson Distribution
- Geometric Distribution

Prob. Distribution of Discrete r.v.

Poisson Distribution – Notation: $X \sim \text{Pois}(\mu)$

- Let r.v. X be the number of “successes” in a given time interval where X is a non-negative integer

Prob. Distribution of Discrete r.v.

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- The r.v. X is a Poisson r.v. with probability distribution given by

$$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$$

where μ is the average number of successes and e is a constant = 2.71828...

Prob. Distribution of Discrete r.v.

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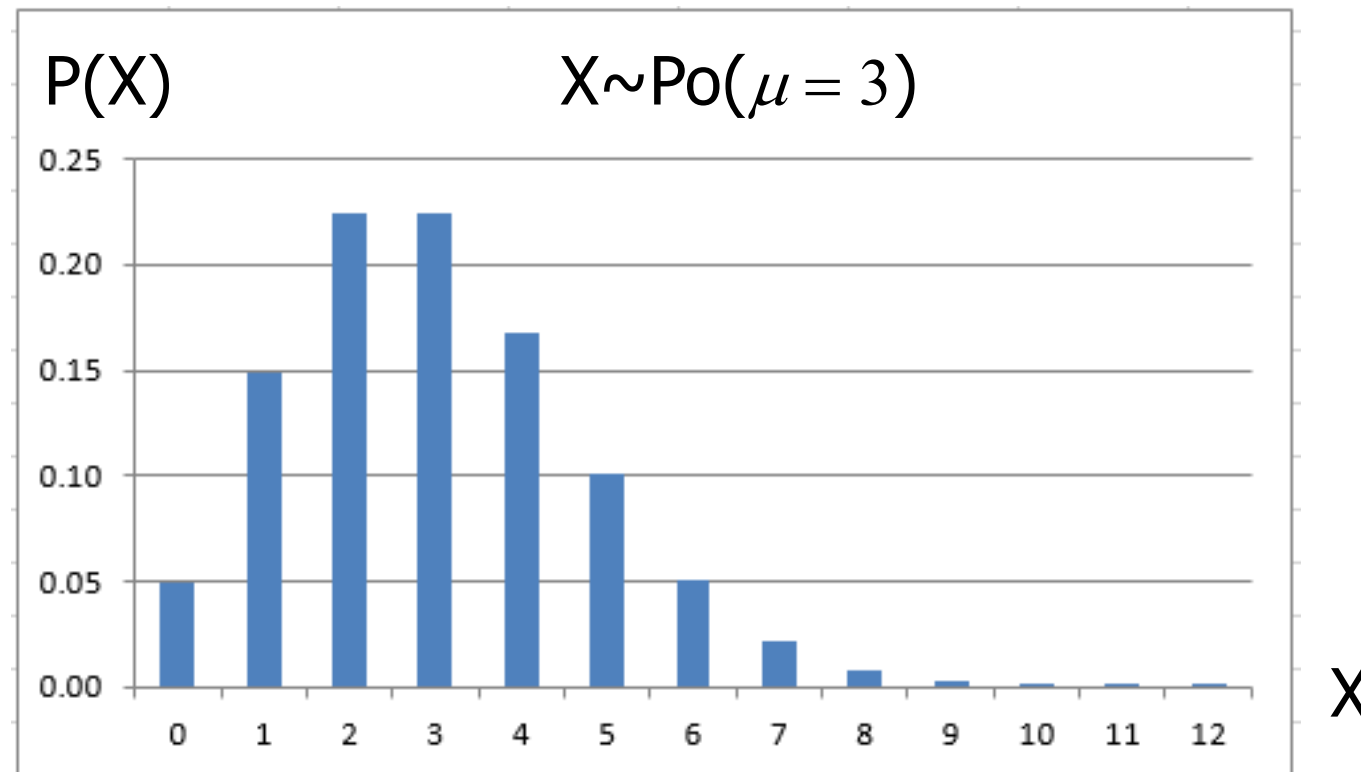
- The expected value and the variance of X are both equal to μ .

Prob. Distribution of Discrete r.v.

Poisson Distribution – $X \sim \text{Pois}(\mu)$

$$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$$

Example:



Prob. Distribution of Discrete r.v.

Eg:

A salesman sells on average 3 iphones per day.
The number of iphones sold per day is a Poisson
r.v.

- (i) Calculate the probability that in a given day
he will sell some iphones.

Prob. Distribution of Discrete r.v.

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- (i) Calculate the probability that in a given day
he will sell some iphones.

$$\begin{aligned} P(\text{some iphones}) &= 1 - P(\text{no iphone}) \\ &= 1 - \frac{e^{-3} 3^0}{0!} \\ &= 0.950 \text{ (to 3.d.p.)} \end{aligned}$$

Prob. Distribution of Discrete r.v.

Eg:

A salesman sells on average 3 iphones per day.
The number of iphones sold per day is a Poisson
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- (ii) Given that there are 8 working hours per day, what is the probability that in a given hour he will sell one iphone?

Prob. Distribution of Discrete r.v.

Eg:

A salesman sells on average 3 iphones per day.
The number of iphones sold per day is a Poisson r.v.

- (ii) Given that there are 8 working hours per day, what is the probability that in a given hour he will sell one iphone?

Avg no. of iphones sold/hour = $3/8 = 0.375$

On a given hour, $P(X=1) = \frac{e^{-0.375} 0.375^1}{1!}$

Example: A shop sells a certain product of which the weekly demand is a Poisson variable with mean 3.

On 1st day of each month, the stocks are replenished.

Obtain the minimum number of the product in stock on 1st day of a month so that the shop is at least 50% sure of being able to meet the demands for the month.

Prob. Distribution of Discrete r.v.

Poisson Distribution – Approx to Binomial Dist

For Binomial Distribution $X \sim B(n, p)$ with large n :
i.e. $n \rightarrow \infty$, $p \rightarrow 0$ and $np \rightarrow$ a constant μ

Prob. Distribution of Discrete r.v.

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Then it can be shown that :

$$\frac{n!}{x! (n-x)!} p^x (1-p)^{n-x} \rightarrow \frac{e^{-\mu} \mu^x}{x!}$$

Prob. Distribution of Discrete r.v.

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$$\frac{n!}{x! (n-x)!} p^x (1-p)^{n-x} \rightarrow \frac{e^{-\mu} \mu^x}{x!}$$

$\therefore X \sim B(n, p) \approx \text{Pois}(\mu)$, where $\mu = np$, we have:

$$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$$

Prob. Distribution of Discrete r.v.

Poisson Distribution – Approx to Binomial Dist

Conditions for which the probability of $X \sim B(n, p)$ can be approximated by the probability $X \sim \text{Pois}(\mu)$:

- large n ($n \geq 100$)
- small p ($p \leq 0.01$)
- constant $\mu = np$ ($\mu \leq 20$)

Intuitive Explanation:

To obtain the mean and variance of the Poisson distribution, consider the binomial distribution under the following conditions:

$n \rightarrow \infty$, $p \rightarrow 0$ and $np \rightarrow \text{a constant } \mu$

The mean and variance of the binomial distribution are given by:

$$E[X] = np$$

$$\text{Var}[X] = npq$$

Since $q = 1 - p$, and as $p \rightarrow 0$, $q \rightarrow 1$ we have:

$$E[X] = \mu \text{ and } \text{Var}[X] = npq = \mu$$

Example: Suppose in a production line, 1 in 200 LED produced is defective. A random sample of 1000 LEDs are selected. What is the probability that at most 2 of them are defective?

Let X = no. of defective LED

p = Prob that a LED is defective = $1/200 = 0.005$

$np = 5$

$n > 100$, $p < 0.01$ and $np < 20$

use Poisson Dist to approx. Binomial Dist

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= e^{-5} + 5e^{-5} + \frac{5^2 e^{-5}}{2!} = 0.125 \text{ (to 3.d.p.)}$$

Prob. Distribution of Discrete r.v.

Special Probability Distributions

- Binomial Distribution
- Poisson Distribution
- Geometric Distribution

Prob. Distribution of Discrete r.v.

Geometric Distribution – Notation: $X \sim G(p)$

- Consider a sequence of independent Bernoulli trials, where each trial is an "experiment" with exactly 2 possible outcomes, "success" and "failure".
- Assume that the probability of success (S) is the same for all trials, $P(S) = p$, and the probability of failure $P(F) = 1 - p$.

Prob. Distribution of Discrete r.v.

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- Each experiment is performed consecutively until the first success is obtained.
- Let X = no. of trials. Hence there are $X - 1$ failures before the 1st success is obtained.

Prob. Distribution of Discrete r.v.

Hence X is a random variable with a Geometric distribution $X \sim G(p)$:

$$P(X = x) = (1 - p)^{x-1} p$$

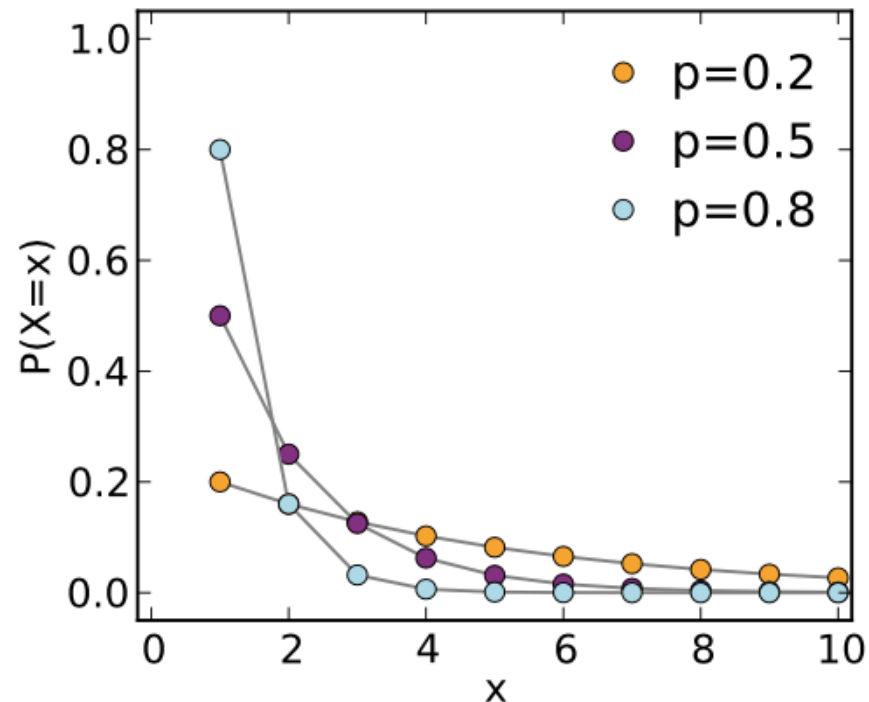
Expected value:

$$E[X] = \frac{1}{p}$$

Variance:

$$\text{Var}[X] = \frac{1 - p}{p^2}$$

Examples:



Example:

In a data transmission, the probability that a data frame is received in error is 0.1. If a data frame is received in error, a retransmission will be requested.

- a) What is probability that a data frame will be successfully received in no more than 3 transmissions?
- b) Determine the average number of transmissions per frame.

Let $X = \#$ of data frames transmitted to get the 1st success

$\therefore X \sim G(p = 0.9)$

$$\begin{aligned} \text{a) } P(X \leq 3) &= P(X=1) + P(X=2) + P(X=3) \\ &= 0.9 + 0.1 * 0.9 + 0.1^2 * 0.9 = 0.999 \end{aligned}$$

$$\text{b) } E[X] = 1/p = 1.111 \text{ (to 3.d.p.)}$$

Prob. Distribution of Discrete r.v.

Special Probability Distributions

- Binomial Distribution
- Poisson Distribution
- Geometric Distribution

Distribution	Probability	Mean	Variance
$X \sim B(n, p)$	$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$	np	$np(1 - p)$
$X \sim \text{Pois}(\mu)$	$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$	μ	μ
$X \sim G(p)$	$P(X = x) = (1 - p)^{x-1} p$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$

Prob. Distribution of Continuous r.v.

- Introduction**
- Special Probability Distributions**
 - Uniform Distribution
 - Exponential Distribution
 - Normal Distribution

Prob. Distribution of Continuous r.v.

A continuous random variable is a variable used for continuous measurements such as distance, weight, and time.

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Probability Density Function (pdf)

For a continuous random variable X , we can describe the probability distribution by some function $f(X)$, such that

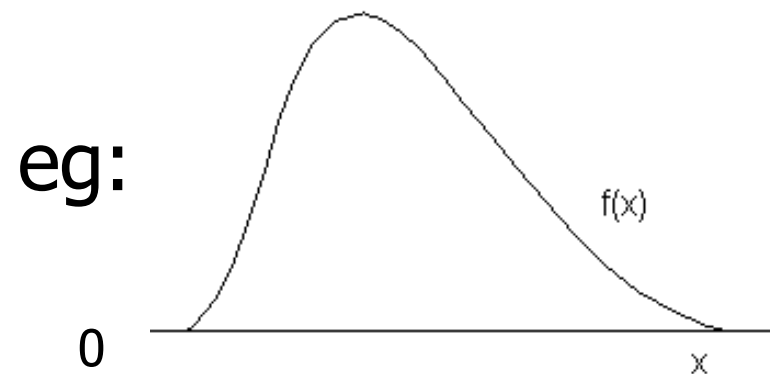
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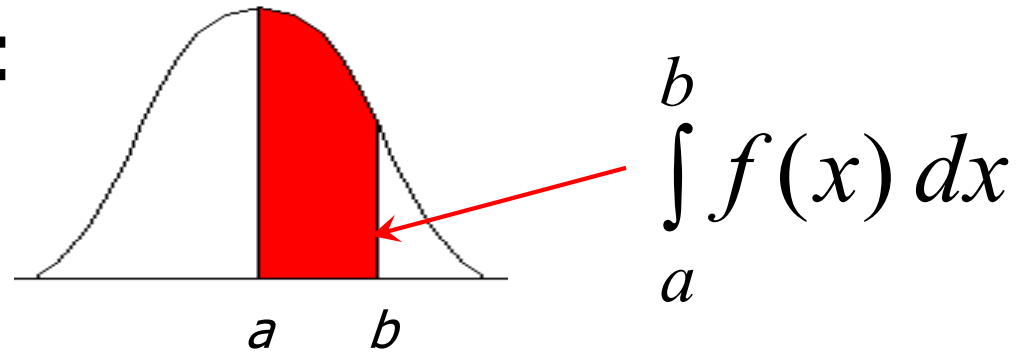
For a continuous random variable X , we can describe the probability distribution by some function $f(X)$, such that

(i) $f(X) \geq 0$ for all X



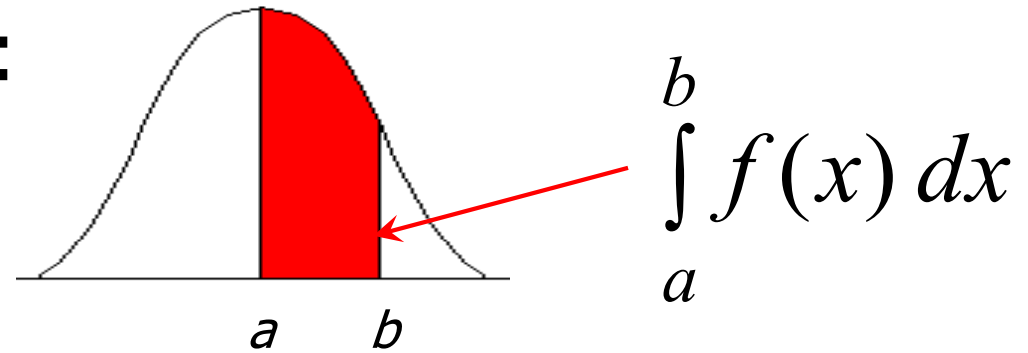
Prob. Distribution of Continuous r.v.

(ii) $P(a < X < b)$ is the area under the curve between a and b :



Prob. Distribution of Continuous r.v.

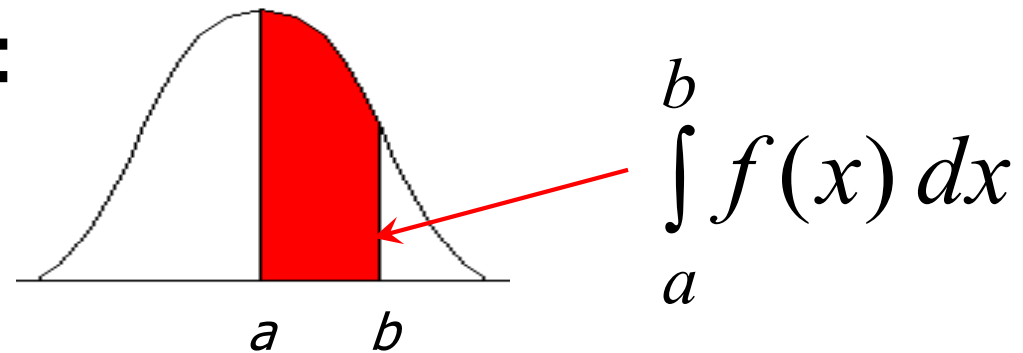
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(iii) Total area under curve = 1.

Prob. Distribution of Continuous r.v.

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(iii) Total area under curve = 1.

The function $f(x)$ is called the probability density function (or pdf) of X .

The probability $P(X < x)$ is known as the Cumulative distribution function (or CDF) of X .

Given that the probability density function of X is:

$$f(x) = \frac{x^3}{4} \text{ where } 0 < x < c ; = 0 \text{ otherwise.}$$

Find the value of the constant c .



Prob. Distribution of Continuous r.v.

For a continuous random variable X the probability that X equals to an exact value is zero.

Eg: for the length of a bolt

$$P(X = 1.999965700 \text{ cm}) = 0$$

i.e., $P(X = x) = 0$.

Prob. Distribution of Continuous r.v.

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Eg: for the length of a bolt

$$P(X = 1.999965700 \text{ cm}) = 0$$

i.e., $P(X = x) = 0$.

Hence, for continuous random variables, probabilities are always associated with a range of values.

Eg: $P(X \leq 2.00 \text{ cm})$ or $P(X > 1.95 \text{ cm})$.

Prob. Distribution of Continuous r.v.

The expected value of a continuous r.v. is defined

as:
$$E[X] = \int_{x=-\infty}^{x=\infty} x f(x) dx = \mu$$

For discrete r.v.

$$E(X) = \sum_{i=1}^n p_i x_i$$

and its variance is:

$$\begin{aligned} \text{var}[X] &= \int_{x=-\infty}^{x=\infty} (x - \mu)^2 f(x) dx \\ &= E[X^2] - E[X]^2 \end{aligned}$$

For discrete r.v.

$$\text{Var}[X] = \sum_{i=1}^n (x_i - \mu)^2 p_i$$

where
$$E[X^2] = \int_{x=-\infty}^{x=\infty} x^2 f(x) dx$$

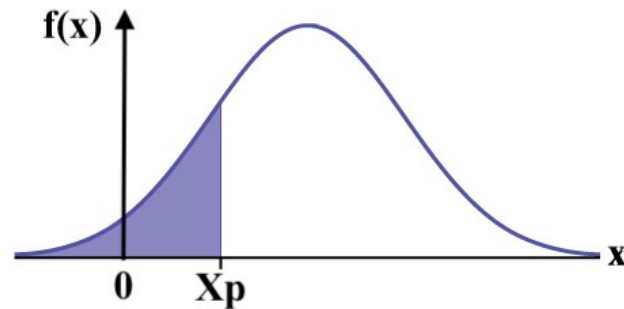
Example:

The pdf of X is given below. Determine $E[X]$.

$$f(x) = \begin{cases} \cos(x), & 0 \leq x \leq \frac{\pi}{2} \\ 0, & \text{otherwise} \end{cases}$$

Prob. Distribution of Continuous r.v.

How do we determine the p^{th} percentile from the probability distribution of a continuous r.v.?



The p^{th} percentile is the value X_p such that the probability that $x < X_p$ is p percent.

$$\int_{-\infty}^{X_p} f(x) dx = \frac{p}{100}$$

$$p^{\text{th}} \text{ percentile} = X_p$$

Wooclap Ex

Given that the probability density function of x is

$$f(x) = e^{-x}, \text{ where } x > 0$$

determine the 25th percentile.



Prob. Distribution of Continuous r.v.

- Introduction

- **Special Probability Distributions**

- Uniform Distribution
- Exponential Distribution
- Normal Distribution

Prob. Distribution of Continuous r.v.

Uniform Distribution

One example of a pdf for a continuous r.v. is the uniform continuous distribution.

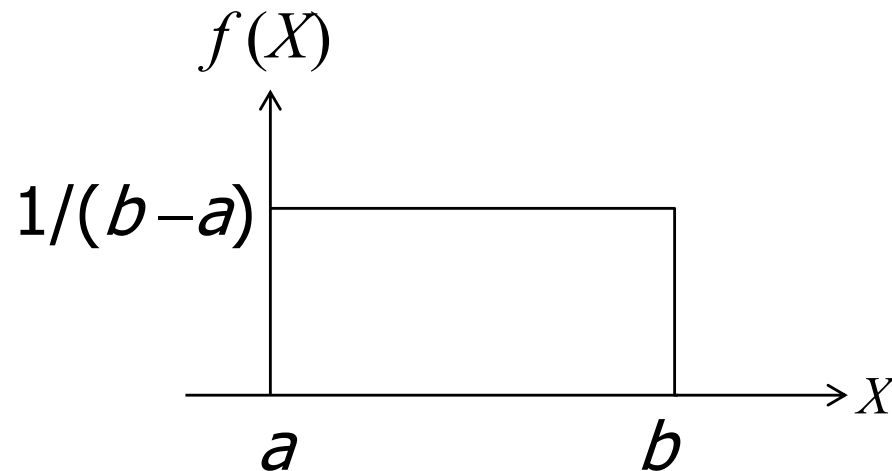
X can take any real value between a and b with uniform probability over this interval.

Notation: $X \sim U(a, b)$.

Total area = 1

Length = $b - a$

Height = $1/(b - a)$

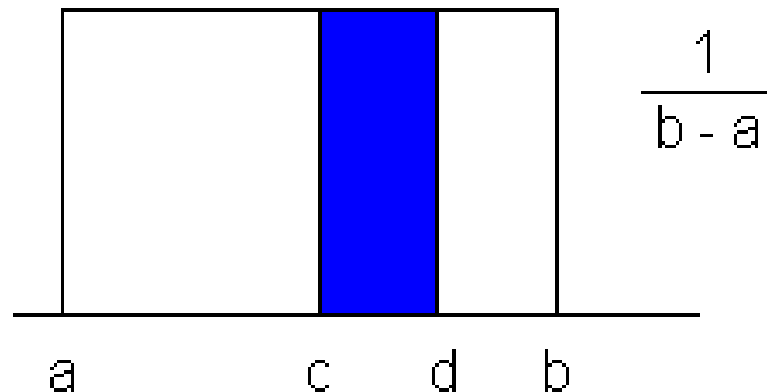


Prob. Distribution of Continuous r.v.

Thus the pdf is $f(X) = \frac{1}{b-a}$ for $a \leq X \leq b$
 $= 0$, otherwise

For any values c and d between a and b

$$P(c < X < d) = \frac{1}{b-a} (d - c)$$



Exercise: Obtain the CDF.

Obtain the expressions for $E[X]$ and $\text{Var}[X]$ of r.v. $X \sim U(a,b)$.

Prob. Distribution of Continuous r.v.

Exponential Distribution

If the number of arrivals during an interval is Poisson distributed, then the interarrival times x are exponentially distributed.

Prob. Distribution of Continuous r.v.

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The pdf of x is:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

where λ is the mean arrival rate and the mean interarrival time is $1/\lambda$.

Prob. Distribution of Continuous r.v.

Exponential Distribution

If the number of arrivals during an interval is Poisson distributed, then the interarrival times x are exponentially distributed.

The pdf of x is:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

where λ is the mean arrival rate and the mean interarrival time is $1/\lambda$.

Notation: $X \sim \text{Exp}(\lambda)$

If the number of arrivals in an interval is Poisson distributed, we show that the interarrival times are exponentially distributed.

Prob. Distribution of Continuous r.v.

Exponential Distribution – examples:

Pdf:

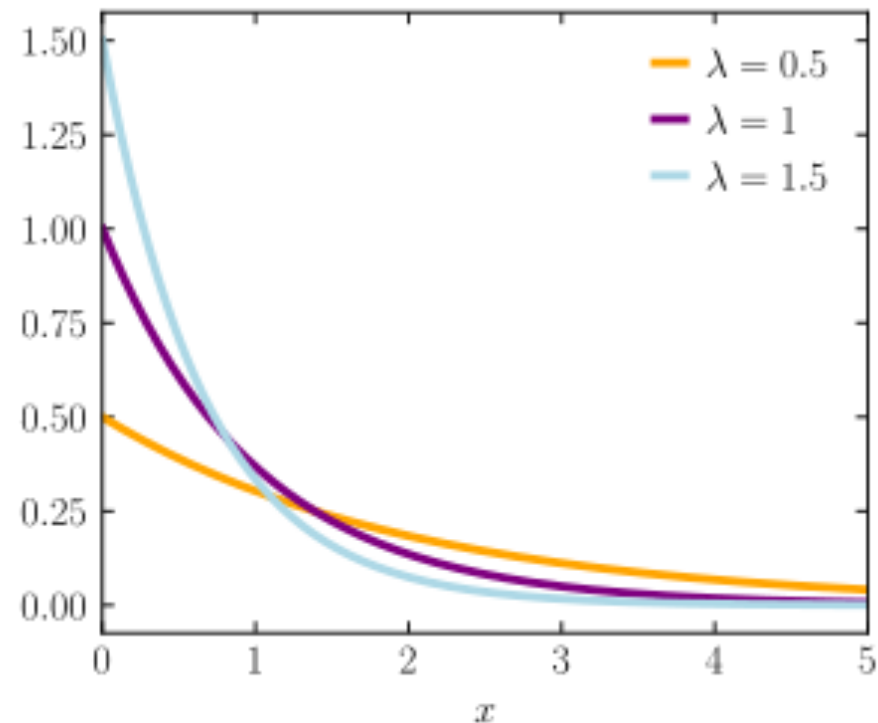
$$f(x) = \lambda e^{-\lambda x}$$

Expected value:

$$E[X] = \frac{1}{\lambda}$$

Variance:

$$\text{Var}[X] = \frac{1}{\lambda^2}$$



Eg: Given that an average of number of customers walk into a shop per hour is 20 and the time between arrivals is exponentially distributed. After a customer arrives, what is the probability that the next customer will arrive within 1 minute?

Prob. Distribution of Continuous r.v.

Normal Distribution

Also known as the Gaussian distribution.

Its bell-shaped pdf depends on μ and σ , and is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{where } -\infty < x < \infty$$

Prob. Distribution of Continuous r.v.

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$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{where } -\infty < x < \infty$$

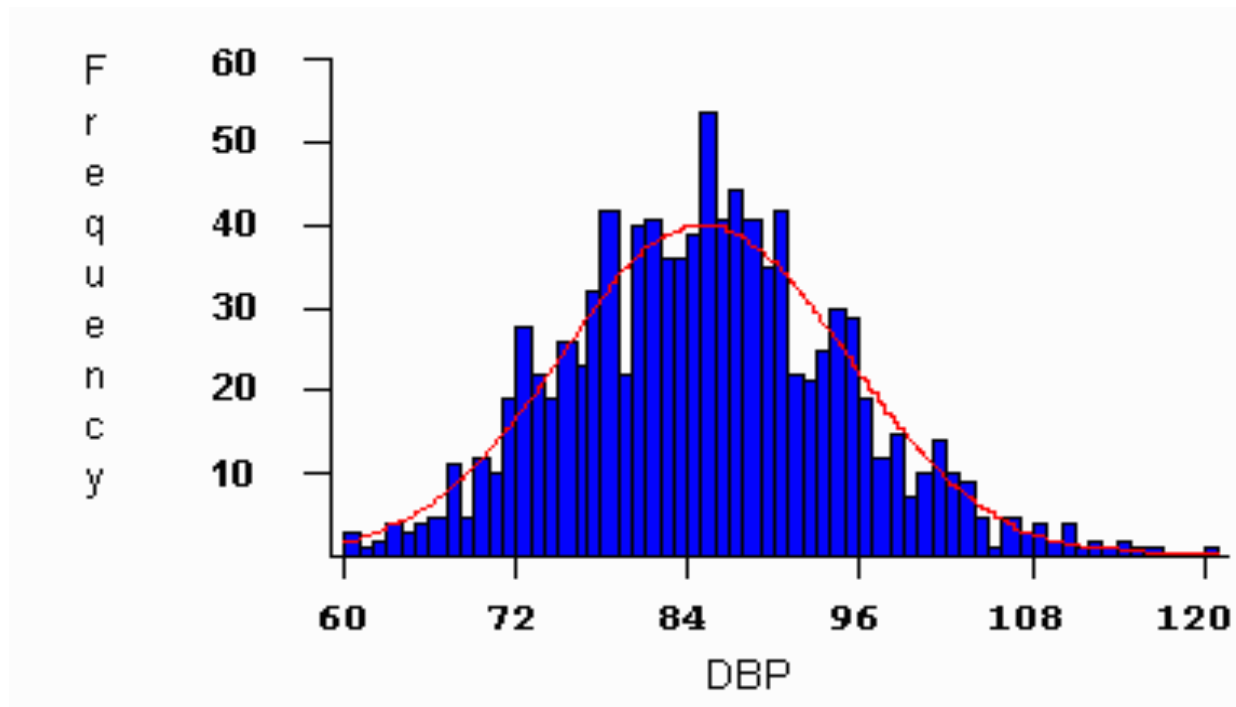
The normal distribution is by convention written as $X \sim N(\mu, \sigma^2)$.

Eg: if $X \sim N(10, 16)$ then this implies that $\mu=10$ and $\sigma^2=16$, and (hence $\sigma=4$).

Prob. Distribution of Continuous r.v.

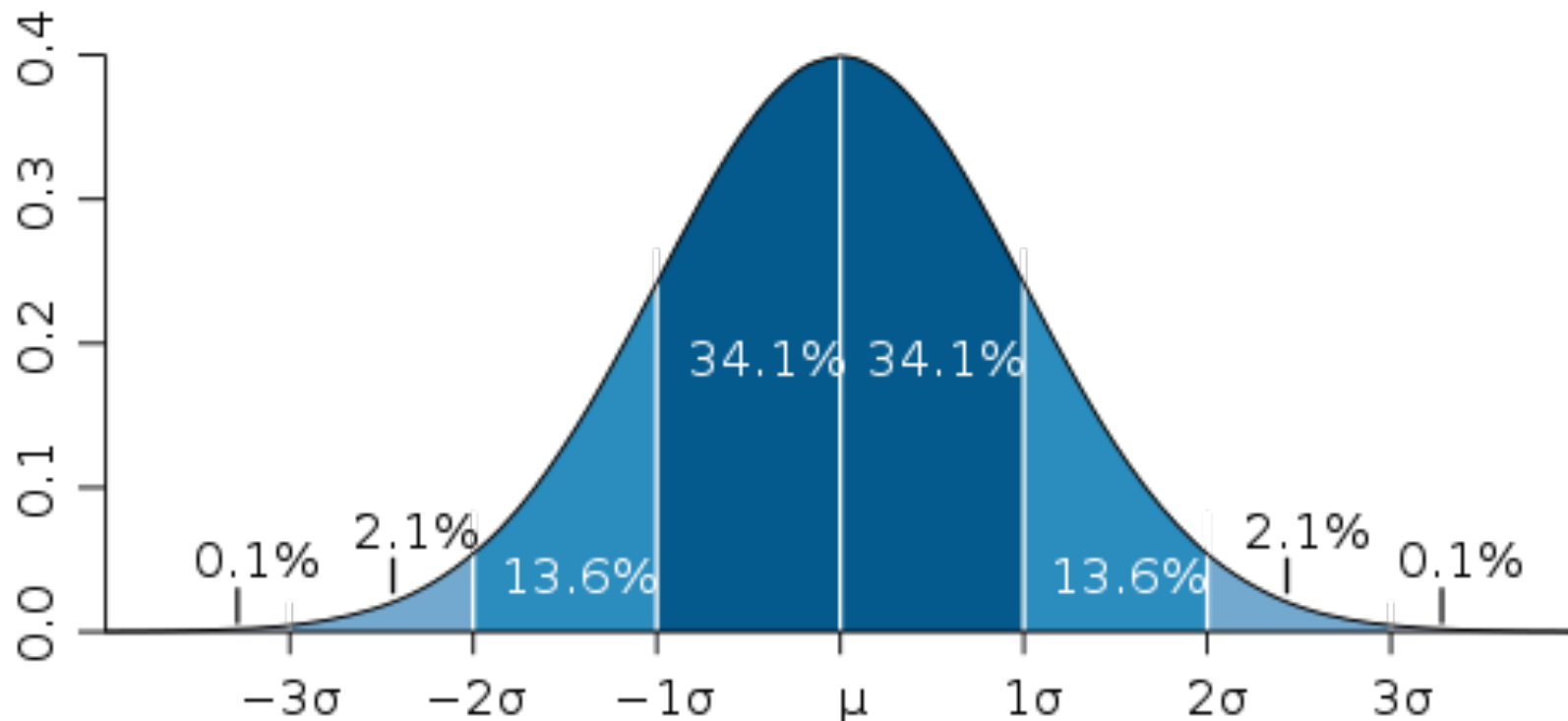
Eg: The distribution of blood pressure in patients can be approximated as a normal distribution with mean 85 mmHg and standard deviation 20 mmHg.

A histogram of 1,000 observations and the normal curve is shown below.



Prob. Distribution of Continuous r.v.

The normal distribution has about 68% of the observations lying within one standard deviation of the mean, 95% within two standard deviations and 99.7% within 3 standard deviations.

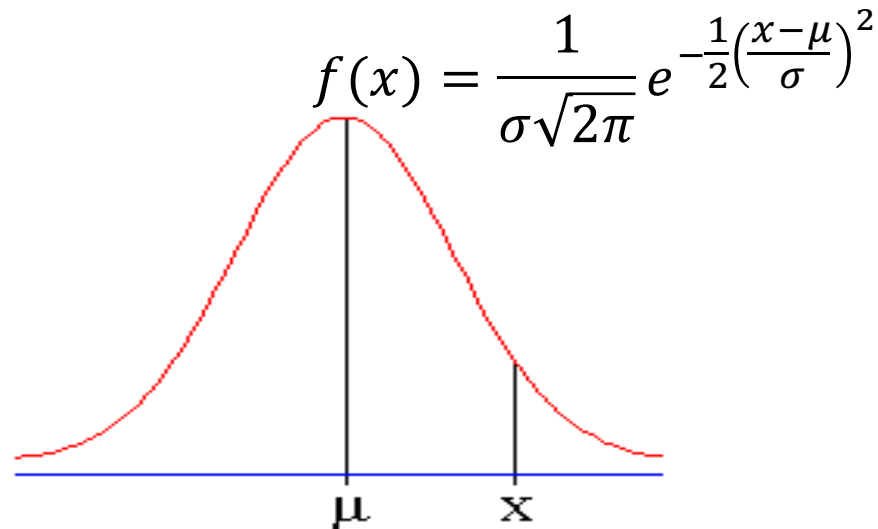


Prob. Distribution of Continuous r.v.

Standard Normal Distribution

To obtain the area under the density curve for a normal distribution, it is necessary to express any value of X in terms of the number of standard deviation units away from μ , i.e.

$$X = \mu + Z\sigma.$$

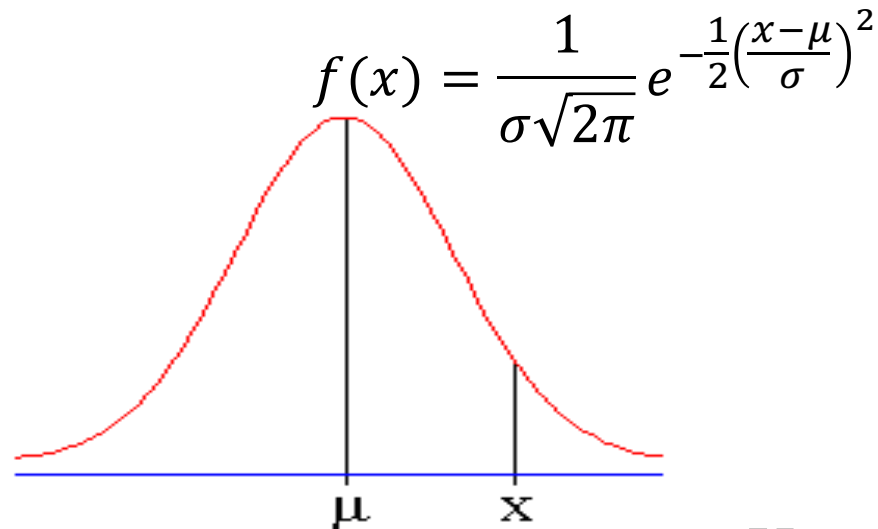


Prob. Distribution of Continuous r.v.

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To obtain the area under the density curve for a normal distribution, it is necessary to express any value of X in terms of the number of standard deviation units away from μ , i.e.

$$X = \mu + Z\sigma.$$



This can be rearranged to give $Z = \frac{X - \mu}{\sigma}$

Prob. Distribution of Continuous r.v.

Since X is a r.v., so is Z .

.. /

Prob. Distribution of Continuous r.v.

Since X is a r.v., so is Z . Using the formulas for functions of random variables we can obtain

$$E[Z] = \frac{1}{\sigma} E[X] - \frac{1}{\sigma} \mu = \frac{1}{\sigma} \mu - \frac{1}{\sigma} \mu = 0$$

$$\text{var}[Z] = \frac{1}{\sigma^2} \text{var}[X] = \frac{\sigma^2}{\sigma^2} = 1$$

Prob. Distribution of Continuous r.v.

Since X is a r.v., so is Z . Using the formulas for functions of random variables we can obtain

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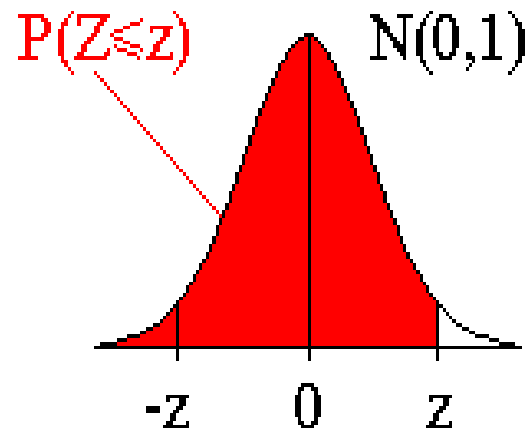
$$\text{var}[Z] = \frac{1}{\sigma^2} \text{var}[X] = \frac{\sigma^2}{\sigma^2} = 1$$

Also it can be shown that Z has a normal distribution. Thus $Z \sim N(0, 1)$.

This is called the standard normal distribution.

Prob. Distribution of Continuous r.v.

Most statistics textbooks include a table of the standard Normal distribution. It tabulates the area under the curve of the $N(0,1)$ density function.



Prob. Distribution of Continuous r.v.

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Other normal tables may give other areas, e.g.

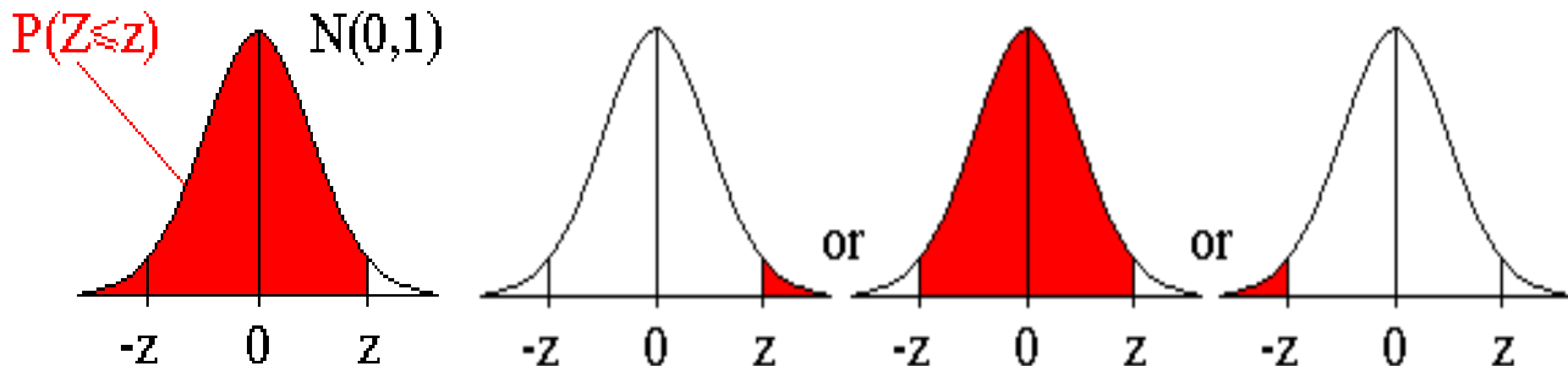


Table of Normal Distribution, $P(Z \leq z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7613	0.7643	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7938	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Eg:

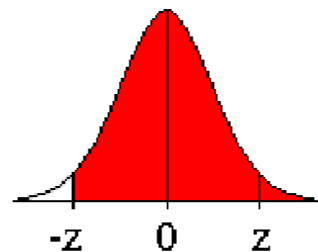
$$P(Z < 0.21) = 0.5832$$

Prob. Distribution of Continuous r.v.

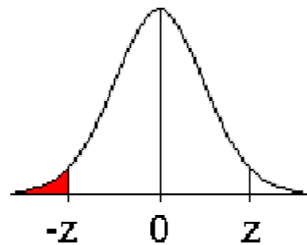
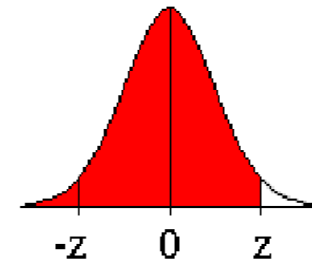
The $P(Z \leq z)$ table gives area under the curve from $-\infty$ to z , where $0 < z < 3.09$

For other values, we perform some manipulations.

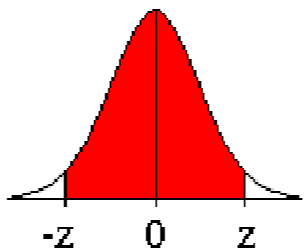
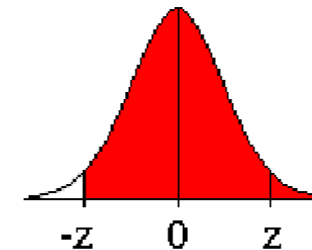
Eg:



$$P(Z \geq -z) = P(Z \leq z)$$



$$P(Z \leq -z) = 1 - P(Z \geq -z)$$



$$P(-z \leq Z \leq z) = 1 - 2P(Z \leq -z)$$

Prob. Distribution of Continuous r.v.

Probabilities for normal distributions other than the standard normal distribution $N(0, 1)$ are obtained by using the formula:

$$Z = \frac{X - \mu}{\sigma}$$

to convert from $X \sim N(\mu, \sigma^2)$ to $Z \sim N(0, 1)$ and then using the table of probabilities for the standard normal distribution $N(0, 1)$.

Prob. Distribution of Continuous r.v.

Normal distribution

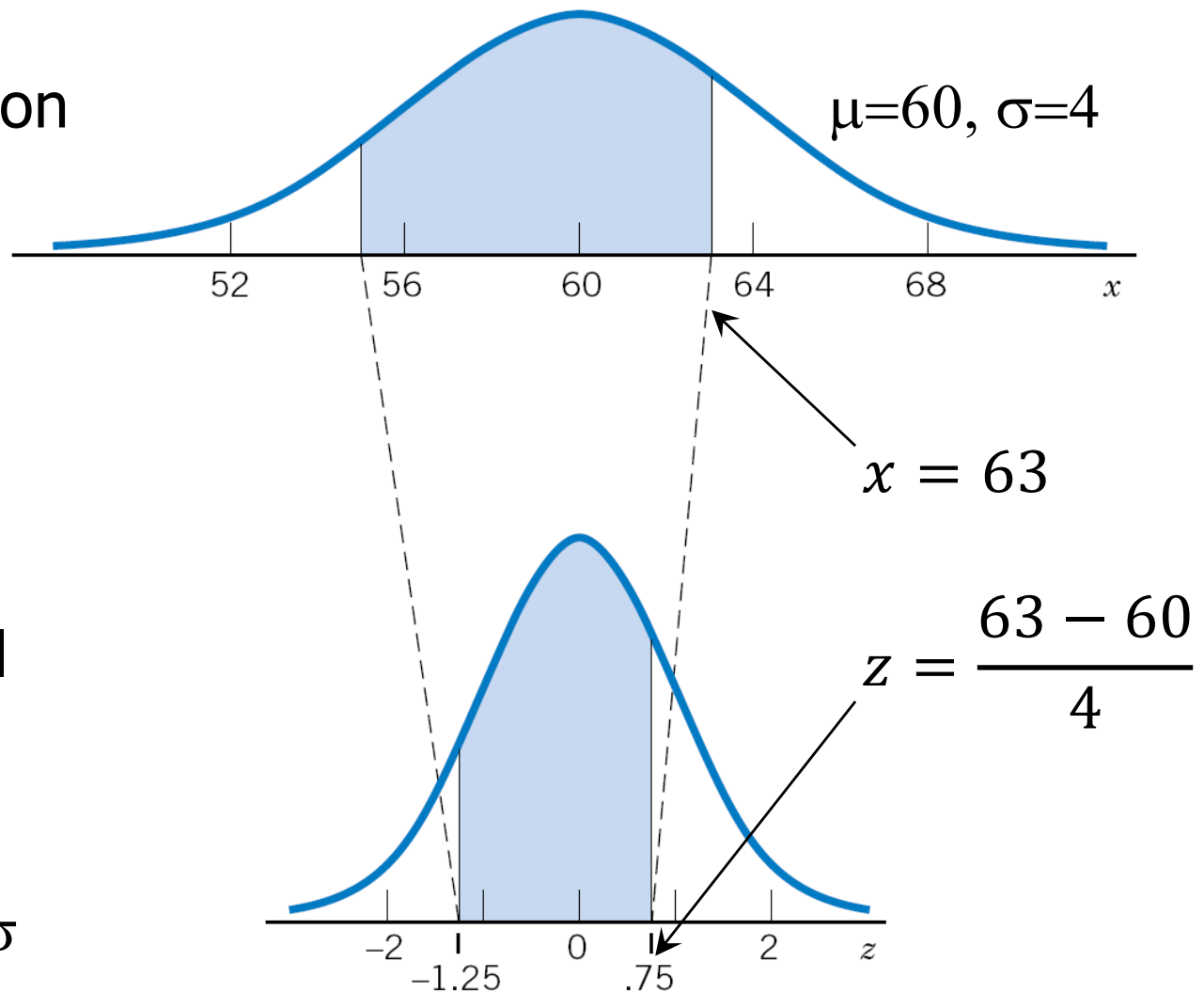
$$X \sim N(\mu, \sigma^2)$$



Standard Normal

$$Z \sim N(0, 1)$$

where $Z = (X - \mu) / \sigma$



Prob. Distribution of Continuous r.v.

Eg: A machine fills bottles with 300 ml of soft drinks. However, the actual quantity filled varies according to the normal distribution with $\mu = 298$ ml and $\sigma = 3$ ml.

What is the probability that an individual bottle contains less than 295 ml?

Prob. Distribution of Continuous r.v.

Solution

Let X denote the quantity in an individual bottle.
Given $X \sim N(298, 3^2)$, we want to find $P(X < 295)$

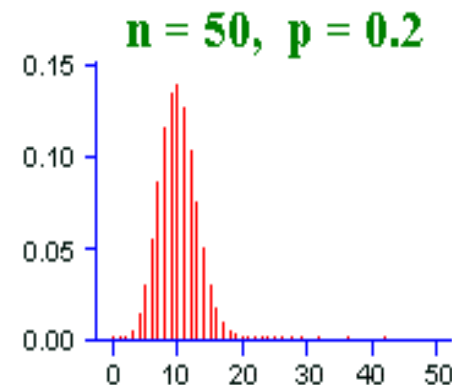
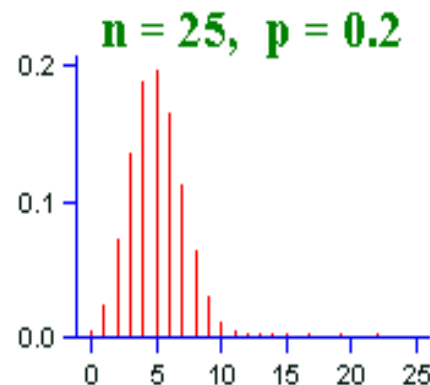
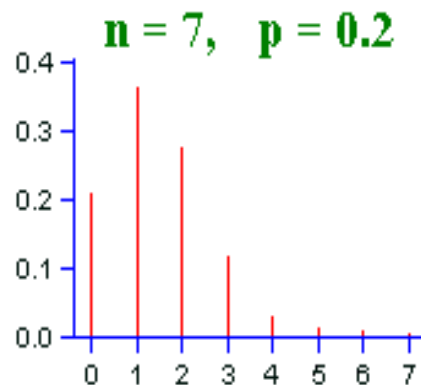
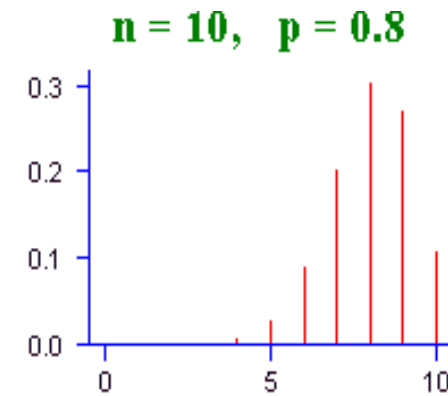
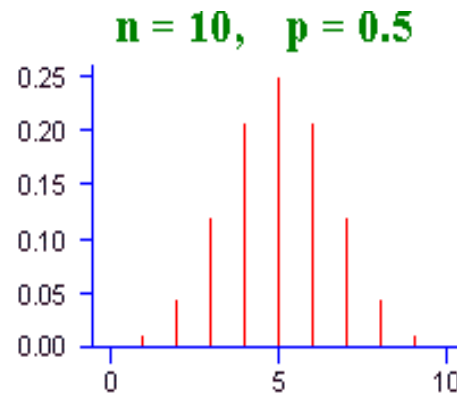
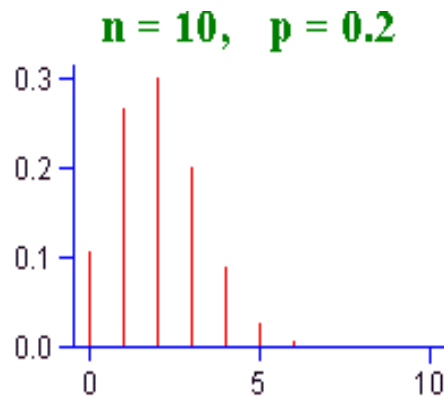
$$\text{If } X=295 \text{ then } Z = \frac{295 - 298}{3} = \frac{-3}{3} = -1$$

$$\begin{aligned} P(X < 295) &= P(Z < -1) = 1 - P(Z < 1) \\ &= 1 - 0.8413 \quad (\text{from tables}) \\ &= 0.1587 \end{aligned}$$

About 16% of bottles would have less than 295 ml.

Approx. to Binomial Distribution

For large n or p near 0.5 the binomial distribution approximately follows the normal distribution.



Approx. to Binomial Distribution

If $X \sim B(n, p)$ where n is large and p not too near 0 or 1, then X can be approximated by a normal distribution with $E(X) = np$ and $\text{var}(X) = npq$, where $q = 1 - p$.

So $Z = \frac{X - np}{\sqrt{npq}}$ is approximately $N(0, 1)$.



This approximation is reasonably good when $np > 5$ and $n(1 - p) > 5$.

Approx. to Binomial Distribution

Continuity Correction and Accuracy

For more accurate values of binomial probabilities, the approximation is improved using the continuity correction. This method considers that each whole number occupies the interval from 0.5 below to 0.5 above it.

Approx. to Binomial Distribution

Continuity Correction and Accuracy

For more accurate values of binomial probabilities, the approximation is improved using the continuity correction. This method considers that each whole number occupies the interval from 0.5 below to 0.5 above it.

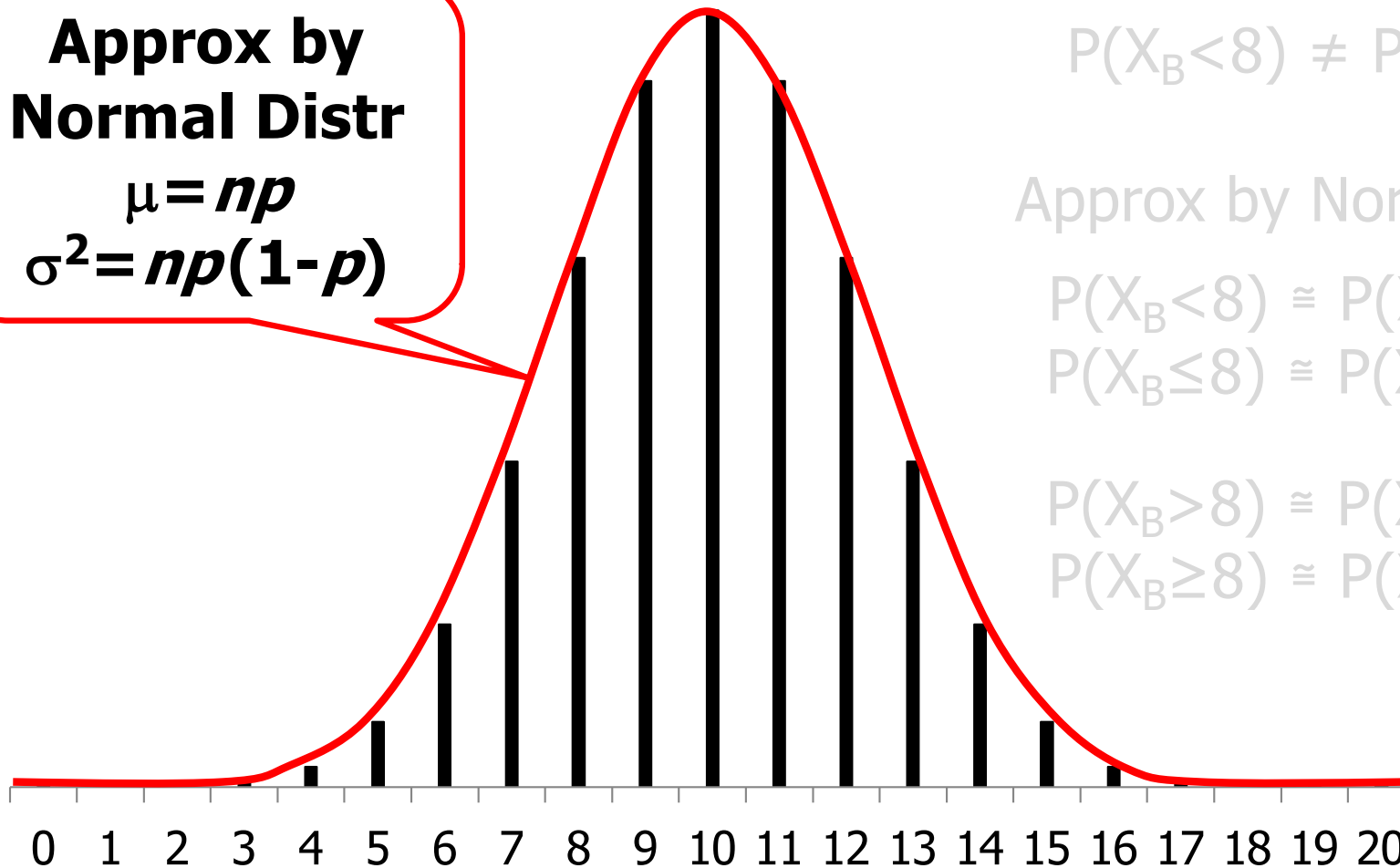
When an outcome X is to be included in the probability calculation, the normal approximation uses either $(X-0.5)$ or $(X+0.5)$.

Approx. to Binomial Distribution

Binomial Distribution $X_B \sim B(n, p)$

**Approx by
Normal Distr**

$$\mu = np$$
$$\sigma^2 = np(1-p)$$



Binomial r.v.

$$P(X_B < 8) \neq P(X_B \leq 8)$$

Approx by Normal distr

$$P(X_B < 8) \approx P(X_N < 7.5)$$

$$P(X_B \leq 8) \approx P(X_N < 8.5)$$

$$P(X_B > 8) \approx P(X_N > 8.5)$$

$$P(X_B \geq 8) \approx P(X_N > 7.5)$$

Approx. to Binomial Distribution

Eg: In a particular faculty 60% of students are men and 40% are women. In a sample of 50 students what is the probability that more than half are women?

- 1st compute the approximate probability with no continuity correction
- 2nd compare the actual result and check the accuracy of the above approximation
- 3rd compute the approximate probability with continuity correction and compare all results

Approx. to Binomial Distribution

We need to find $P(X > 25)$. Note: not $P(X \geq 25)$

No continuity correction

More than $\frac{1}{2}$
are women

$$\text{So } P(X > 25) = P\left(Z > \frac{25 - 20}{\sqrt{12}} = 1.44\right)$$

$$= 1 - P(Z \leq 1.44)$$

$$= 1 - 0.9251 = 0.075$$

The exact ans. calculated from binomial probabilities

$$\begin{aligned} \text{is: } P(X > 25) &= P(X = 26) + P(X = 27) + \dots + P(X = 50) \\ &= 0.0573 \end{aligned}$$

Approx. to Binomial Distribution

The approximate probability, using the continuity correction, is

$$\begin{aligned}P(X > 25) &= P\left(Z > \frac{25.5 - 20}{\sqrt{12}}\right) \\&= P(Z > 1.59) \\&= 1 - P(Z \leq 1.59) \\&= 0.0559\end{aligned}$$

Continuity correction

Compare to 0.0573

which is a more accurate approximation than the original approximation (without correction) of 0.075.

Approx. to Binomial Distribution

The value 25.5 was chosen as the outcome 25 was not to be included but the outcomes 26, 27, ..., 50 were to be included in the calculation.

Similarly, if the example required the probability that less than 18 students were women, the continuity correction would require the calculation

$$P(X < 18) = P\left(Z < \frac{17.5 - 20}{\sqrt{12}}\right)$$

Summary

- Random Variables
- Expected Values and Variance
- Discrete Distributions
 - 1) Binomial
 - 2) Poisson
 - 3) Geometric
- Continuous Distributions
 - 1) Uniform
 - 2) Exponential
 - 3) Normal and Standard Normal
- Normal Approximation to Binomial

Distribution	Probability	Mean	Variance
$X \sim B(n, p)$	$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$	np	$np(1 - p)$
$X \sim \text{Pois}(\mu)$	$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$	μ	μ
$X \sim G(p)$	$P(X = x) = (1 - p)^{x-1} p$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$
$X \sim U(a, b)$	$f(x) = \begin{cases} \frac{1}{b - a} & a < x < b \\ 0 & \text{otherwise} \end{cases}$	$\frac{a + b}{2}$	$\frac{(a - b)^2}{12}$
$X \sim \text{Exp}(\lambda)$	$f(x) = \lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
$X \sim N(\mu, \sigma^2)$	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	μ	σ^2